Software Requirements Specification

For

Mittal Yarns

Version 1.0 approved

Prepared by

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Revision History

Name	Date	Reason For Changes	Version
Mittal Yarns	10-03-2026	Initial Version	Initial Draft
Mittal Yarns	18-03-2026	Change in Security Function	Version 1.0

1. Introduction

1.1 Purpose

This document defines the software requirements for the Mittal Spinners Web System, a corporate website and internal management platform designed to represent Mittal Spinners digitally and streamline factory operations.

The system replaces the current manual processes for order management and worker attendance tracking. It includes a public-facing corporate website for product display and customer inquiries, an attendance tracking module for worker management, and an order management module for production and dispatch monitoring.

1.2 Document Conventions

- 'Shall' indicates a mandatory requirement.
- 'Should' indicates a recommended requirement.
- 'May' indicates an optional requirement.
- Requirements are uniquely identified.
- All requirements are atomic, testable, and verifiable.
- Priority levels: High, Medium, Low.

1.3 Intended Audience and Reading Suggestions

This document is intended for the Developers, testers, project managers, and stakeholders

Audience	Relevant Sections
Software Developers	Sections 3, 4, 5 – functional requirements and system interfaces
QA / Test Engineers	Sections 4, 5 – testable stimulus/response sequences and validation rules
Project Managers	Sections 1, 2, 7 – scope, constraints, and non-functional requirements
Business Stakeholders (Mittal Spinners)	Sections 1, 2 – product scope and business overview
System Architects	Sections 2, 3, 4 – architecture and interface requirements

1.4 Product Scope

The Mittal Spinners Web System encompasses the following:

- A public corporate website with product showcase, technical specifications, manufacturing process details, and quality control information.
- A customer inquiry system with email notifications to the Admin.
- A secure role-based internal portal for Admin and Supervisor users.
- A multi-level attendance management system with daily marking, approval workflows, and report generation.
- An order tracking system managing order lifecycle from creation to delivery.

Out of Scope (Version 1.0):

- E-commerce / online payment integration.
- Mobile application (Android/iOS).
- Real-time machine or production-line sensor integration.
- Multi-branch or multi-factory support.

1.5 References

- IEEE SRS Standard (IEEE Std 830-1998)
- React Documentation: https://react.dev/
- Vite Documentation: https://vitejs.dev/
- Tailwind CSS Documentation: https://tailwindcss.com/docs
- Node.js Documentation: https://nodejs.org/en/docs
- Express.js Documentation: https://expressjs.com/
- PostgreSQL Documentation: https://www.postgresql.org/docs/
- Prisma ORM Documentation: https://www.prisma.io/docs/

2. Overall Description

2.1 Product Perspective

The Mittal Spinners Web System is a standalone web-based platform. It has no dependency on existing legacy software. The system is structured into two primary components: a public-facing website and a password-protected internal management portal. Both components share a single database and are hosted on a common web server.

The public website serves as the digital storefront for Mittal Spinners, enabling customers to view the product catalogue, understand the manufacturing process, and submit inquiries. The internal portal enables Admin and Supervisor roles to manage workers, mark attendance, handle orders, and generate reports.

2.2 Product Functions

- Public corporate website with product catalogue and inquiry form.
- Implement a secure authentication system for Admin and Supervisor users.
- Worker and department management.
- Multiple level attendance marking system.
- Order creation, stage tracking, and dashboard monitoring.
- Inquiry management with order conversion facility.
- Notification system with email notifications.
- Download facility for generated reports in PDF and Excel formats.
- Admin CMS for managing homepage banners and company information.

2.3 User Classes and Characteristics

User Class	Description	Technical Level	Access Scope
Admin (Owner)	Full system access: manage products, orders, workers, and all modules	Low-to-medium	Full
Supervisor	Mark daily attendance, view orders for assigned department	Low	Attendance + Orders (read)
Customer (Public)	Browse products, submit inquiry forms	None	Public website only

2.4 Operating Environment

- Server OS: Windows 10/11 or macOS (latest stable versions).
- Backend Runtime: Node.js (v18 or higher) with Express.js framework.
- Frontend: React (with Vite) and Tailwind CSS.
- Database: PostgreSQL (v15 or higher) with Prisma ORM.
- Client: Any modern web browser (Chrome, Firefox, Edge, Safari).
- Display: Minimum 1280×720 resolution for internal portal; fully responsive for public website.
- Network: Standard internet connection required for both public website and internal portal access.

2.5 Design and Implementation Constraints

- The website should be fully responsive and mobile-friendly for the public-facing side.
- The portal routes should be protected with authentication middleware; otherwise, it should redirect to the login page.
- The page load time should not be more than 3 seconds for the public-facing side.
- The data belongs to Mittal Spinners.
- The development team should not store any business data.
- The entries added to the attendance should be immutable if approved by Admin; changes should be re-approved by Admin

2.6 User Documentation

The system shall be delivered with the following user documentation components to assist end users and administrators:

- A User Manual describing system features, navigation, and step-by-step instructions for performing key tasks such as attendance marking, order management, and report generation.
- Online Help/Guidance integrated within the application to provide contextual assistance and error explanations.
- A Quick Start Guide for initial system setup and basic usage for Admin and Supervisor roles.

- Technical Documentation for developers, including system architecture, API endpoints, and database schema.

2.7 Assumptions and Dependencies

- Mittal Spinners will offer hosting infrastructure prior to the final deployment phase.
- All product images, descriptions, and branding will be provided by the client during the design phase.
- Employees do not have individual logins to the system; attendance is entered by the Supervisor on behalf of the employee.
- Access to an SMTP service for notification delivery is provided to the client.
- One account exists for the Admin during the initial setup.

3. External Interface Requirements

3.1 User Interfaces

- Public Website: Responsive design for Home, About Us, Products, Manufacturing Process, Quality Control, and Contact Us pages. Validated inquiry form and interactive location map.
- Admin Dashboard: Summary cards for active orders, pending inquiries, pending attendance approvals, and recent activity log. Links to all modules.
- Attendance Module: Date selector and department selector for supervisors. Worker list with status dropdown (Present / Absent / Half-day / Leave). Admin view includes approval queue.
- Order Management: Form to create order with customer information, product specifications, order quantity, and expected dispatch date. List view of orders with badges indicating order stage and option to filter orders by status.
- Reports Section: Date range selectors with export buttons to generate attendance report as PDF or Excel and order report as PDF or Excel.
- All destructive or irreversible actions should be confirmed with a dialog box before the action is executed.
- Validation errors should be shown inline next to the field with the error.

3.2 Hardware Interfaces

The system is a web-based application and does not require direct interaction with specialized hardware components. However, it shall interface with standard computing devices as follows:

- The system shall function on user devices, which may include desktop computers and laptops running Windows or macOS.
- The system shall allow interaction through the usage of standard input devices, which may include a keyboard and mouse or trackpad.
- The system shall allow the display of output through the usage of standard display devices with a resolution of at least 1280×720 pixels.
- The system shall allow interaction through the usage of standard networking devices using TCP/IP protocols for internet access.
- The server running the application shall function on standard devices capable of running Node.js and PostgreSQL database operations.

No specialized hardware interfaces or direct low-level hardware control are required for the system.

3.3 Software Interfaces

- Database: MySQL 8.0 via Laravel Eloquent ORM. Connection configured via encrypted .env file.
- Email: SMTP integration (configurable: Gmail SMTP or custom SMTP relay) for inquiry notifications, attendance submission alerts, and order status updates.
- PDF Export: DomPDF (PHP library) for generating attendance and order reports. No external office suite required at runtime.
- Excel Export: Laravel Excel (Maatwebsite) for generating downloadable .xlsx reports.
- Maps: Google Maps Embed API for the interactive location map on the Contact page.

3.4 Communications Interfaces

The system shall support the following communication interfaces and protocols for data exchange and user interaction:

- The system should operate on the HTTP/HTTPS protocol. HTTPS should be used for secure communication between the client and the server.
- The client-side application (React frontend) should communicate with the backend application (Node.js + Express) using RESTful APIs, with the data being sent in the JSON format.
- The system should be able to operate through standard web browsers like Chrome, Firefox, Edge, Safari, over the internet.
- The system should be able to operate with an SMTP-based email service for notifications like acknowledgment of inquiries, updates on orders, etc.
- All communication should be encrypted using the SSL/TLS protocol for secure communication when handling sensitive data.
- The system should be able to operate with authentication mechanisms like token-based authentication for secure API communication.
- The system should be able to operate at standard broadband data transfer rates with efficient data exchange with minimal latency.

No additional communication protocols such as FTP or real-time socket communication are required for the current system.

4. System Features

4.1 Public Website

4.1.1 Description and Priority

The Public Website module provides an interface for external users to access company information, view services, and submit customer inquiries through a contact form. It serves as the primary entry point for potential customers and enables initial interaction with the organization.

Priority: High

Priority Ratings:

- Benefit: 9
- Penalty: 7
- Cost: 5
- Risk: 4

4.1.2 Stimulus/Response Sequences

Stimulus (User Action)	System Response
Visitor navigates to the website	The System serves the homepage with a hero section, product highlights, and navigation within 3 seconds.
Visitor submits inquiry form with valid data	The System stores inquiry in the database and sends an email notification to Admin. The visitor sees a confirmation message.
Visitor submits inquiry form with invalid data	The System displays inline validation errors adjacent to the offending fields. No data is stored.
Admin adds a new product	The System stores product details and images. Product appears immediately in the public catalogue.
Admin deletes a product	System removes the product from the catalogue after confirmation dialog.

4.1.3 Functional Requirements

REQ-WEB-1: The system shall provide a public corporate website with company profile information, product catalogue with technical specification details, information on manufacturing processes, and quality control information. The website shall be responsive to all mobile device sizes and desktop sizes.

REQ-WEB-2: The administrator will be able to add, modify, and remove product descriptions, pictures, and downloadable catalogs (PDF) from the system. When a change is saved, it will instantly appear on the public website.

REQ-WEB-3: The system must have a validated customer inquiry form that: (a) verifies all required fields prior to submission; (b) saves the inquiry in the database with a timestamp; and (c) notifies the administrator via email. Name (VARCHAR(100)), Phone (10-digit), Email (valid format), and Message (minimum 20 characters) are required fields.

REQ-WEB-4: On the Contact page, the system will show the company's address, phone numbers, email addresses, and an interactive Google Maps location.

REQ-WEB-5: System shall allow the Admin to update homepage banner images and company information (About Us, mission statement) without developer assistance via a CMS interface.

4.2 Authentication & Roles

4.2.1 Description and Priority

The Authentication & Roles module is responsible for managing user logins, logouts, and access control within the system. This module allows only authorized users to access the internal portal. It provides access control for Admins and Supervisors.

Priority: High

Priority Ratings:

- Benefit: 9
- Penalty: 9
- Cost: 6
- Risk: 7

4.2.2 Stimulus/Response Sequences

Stimulus	System Response
User submits valid credentials	The System authenticates the user, creates a session, and redirects to the role-appropriate dashboard.
User submits invalid credentials	The System displays ERR-AUTH-001; increments failed attempt counter. Account locked after 5 consecutive failures.
Session is idle for 60 minutes	The System logs out the user automatically and redirects to the login page with ERR-AUTH-002.
User requests password reset	The System sends a time-limited reset link to the registered email address.

4.2.3 Functional Requirements

- REQ-AUTH-1: For Admin and Supervisor jobs, the system will offer secure role-based authentication. Bcrypt will be used to hash user passwords. Passwords in plaintext should never be kept or recorded.
- REQ-AUTH-2: System shall enforce session timeout after 60 minutes of inactivity. Users are redirected to the login page and must re-authenticate.
- REQ-AUTH-3: The system will send out an email link to reset the password. After 30 minutes, reset links will expire.
- REQ-AUTH-4: Worker ID, Name, Department, Shift, Employment Category (permanent/contract), and Status (active/inactive) are among the worker master records that the system must keep. Worker data can be added, edited, deactivated, and bulk imported by the administrator using CSV.
- REQ-AUTH-5:The system will restrict login to deactivated accounts and provide a suitable error message.

4.3 Attendance Management

4.3.1 Description and Priority

The Attendance Management module enables Supervisors to record daily attendance for workers and allows Admins to review, approve, and manage attendance records. It ensures accurate tracking of workforce activity and supports report generation for operational analysis.

Priority: High

Priority Ratings:

- Benefit: 9
- Penalty: 8
- Cost: 6
- Risk: 6

4.3.2 Stimulus/Response Sequences

Stimulus	System Response
Supervisor marks attendance for a worker	System records attendance status (Present/Absent/Half-day/Leave) with timestamp and Supervisor ID. Status set to Pending Approval.
Supervisor tries to mark attendance already submitted today	The System rejects ERR-ATT-001 and displays the existing record.
Admin approves attendance entry	System marks the entry as Approved and logs Admin ID and approval timestamp.
Admin edits an attendance entry	The system updates the record, logs the change with old value, new value, Admin ID, and timestamp in the audit log.
Admin generates monthly attendance report	The system computes a per-worker attendance summary and generates downloadable PDF and Excel files.

4.3.3 Functional Requirements

- **REQ-ATT-1:** System shall allow Supervisors to mark daily attendance for workers in their assigned department with statuses: Present, Absent, Half-day, and Leave. Each submission is timestamped and linked to the submitting Supervisor's account.
- **REQ-ATT-2:** System shall prevent duplicate attendance entries for the same worker on the same date. A second submission raises ERR-ATT-001.
- **REQ-ATT-3:** System shall generate monthly and department-wise attendance reports. Reports show per-worker totals for each status type (Present, Absent, Half-day, Leave) and are exportable as PDF and Excel.
- **REQ-ATT-4:** System shall maintain a full attendance history with an audit log recording: record ID, worker ID, date, status, submitted by (Supervisor ID), submitted at (timestamp), approved by (Admin ID or null), approved at (timestamp or null), last modified by, last modified at.
- **REQ-ATT-5:** System shall allow the Admin to review, approve, and edit attendance entries. All edits are captured in the audit log with the original and updated values. Approved entries may only be modified by Admin.

4.4 Order Management & Tracking

4.4.1 Description and Priority

The Order Management & Tracking module allows Admins to create, update, and monitor customer orders through defined production stages. It ensures systematic workflow progression, real-time tracking, and visibility into order status from creation to delivery.

Priority: High

Priority Ratings:

- Benefit: 9
- Penalty: 8
- Cost: 7
- Risk: 6

4.4.2 Stimulus/Response Sequences

Stimulus	System Response
Admin creates a new order	The System stores orders with customer details, product specifications, quantity, expected dispatch date, and sets status to 'New'. Assigns a unique order reference.
Admin advances order to next stage	The System validates the stage transition, updates status, logs the change with timestamp and Admin ID.
Admin attempts an invalid stage transition	The System rejects ERR-ORD-002 and displays allowed next stages.
Admin views the orders dashboard	The System displays cards/counts for active, completed, and overdue orders with recent history logs.

4.4.3 Functional Requirements

- **REQ-ORD-1:** System shall allow the Admin to create and manage orders with: customer name, contact details, product type, yarn specifications (count, ply, blend), quantity (kg), expected dispatch date, and notes.
- **REQ-ORD-2:** System shall track order progress through the following defined stages in sequence: New → In Production → Quality Check → Packed → Dispatched → Delivered. No stage may be skipped. Invalid transitions are rejected with ERR-ORD-002.
- **REQ-ORD-3:** System shall display a dashboard showing active orders (by stage), completed orders, overdue orders (expected dispatch date passed and not yet Dispatched), and a history log of recent stage transitions.
- **REQ-ORD-4:** System shall send an email notification to the customer (if email is provided) when the order status changes to Dispatched or Delivered.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

5.1.1 **Page Load Time**

- Public website pages shall load within **3 seconds** under normal broadband conditions (≥10 Mbps).
- Internal portal pages (dashboard, attendance list, order list) shall load within **2 seconds** after successful authentication.
- Rationale: Ensures acceptable user experience and reduces abandonment rates.

5.1.2 **Report Generation**

- PDF and Excel report generation shall complete within **10 seconds** for datasets covering up to **one month** of data.
- Rationale: Supports operational efficiency for administrative users.

5.1.3 **Database Query Performance**

- Queries for attendance and order data shall return results within **1 second** for datasets up to **10,000 worker-day records**.
- Rationale: Ensures real-time usability of dashboards and operational tools.

5.1.4 **Concurrent Users**

- The system shall support at least **100 concurrent authenticated users** without degradation beyond the defined response times.
- Rationale: Supports organizational scalability.

5.2 Safety Requirements

The system shall enforce strict security and privacy controls to protect user data and system integrity:

5.2.1 Authentication and Credential Security

- All user passwords shall be hashed using **bcrypt**.
- Plaintext passwords shall **never be stored or logged**.
- Requirement: Minimum password complexity rules shall be enforced.

5.2.2 Access Control

- All internal portal routes shall require authentication via middleware.
- Unauthenticated users shall be redirected to the login page.
- Role-Based Access Control (RBAC) shall be enforcedSupervisors shall not access order management, inquiry management, or report generation.

5.2.3 Application Security

- The system shall be protected against common web vulnerabilities:
  - SQL Injection → prevented via parameterized queries.
  - Cross-Site Scripting (XSS) → prevented via output escaping.
  - Cross-Site Request Forgery (CSRF) → prevented via CSRF tokens.

5.2.4 Configuration Security

- Sensitive credentials (database, SMTP) shall be stored in an **encrypted environment configuration file (.env)** and not in source code.

5.2.5 Session Management

- User sessions shall expire after **30 minutes of inactivity**.
- Rationale: Reduces risk of unauthorized access.

5.3 Security Requirements

The system shall exhibit the following measurable quality characteristics:

5.3.1 Availability

- The system shall achieve ≥ 99% uptime excluding scheduled maintenance.
- Priority: High

5.3.2 Reliability

- The system shall ensure consistent operation with <1% failure rate for critical transactions (attendance submission, order creation).
- Priority: High

5.3.3 Usability

- Users shall be able to complete core tasks (attendance marking, order creation) within 3 clicks or steps after login.
- Preference: Ease of use over ease of learning.

5.3.4 Maintainability

- The codebase shall follow modular architecture (MVC) to allow updates without impacting unrelated modules.
- Metric: New features should be implementable with minimal changes (<20% codebase impact)

5.3.5 Scalability

- The system shall support horizontal scaling of backend services.
- Metric: Performance should degrade by no more than 20% when user load doubles.

5.3.6 Security

- The system shall comply with standard secure coding practices and OWASP guidelines.
- Priority: Critical

5.3.7 Testability

- The system shall support unit and integration testing with at least 70% code coverage.

5.4 Software Quality Attributes

The system shall satisfy the following key quality attributes to ensure performance, usability, and long-term maintainability:

5.4.1 Availability

- The system shall maintain an uptime of **at least 99%**, excluding scheduled maintenance.
- Scheduled maintenance shall be notified at least **24 hours in advance**.
- Priority: High

5.4.2 Reliability

- Critical operations (attendance submission, order creation, approvals) shall have a failure rate of **less than 1%**.
- The system shall allow safe retry of failed operations without data inconsistency.
- Priority: High

5.4.3 Usability

- Users shall be able to complete primary tasks (attendance marking, order creation) within **3–4 steps after login**.
- The system shall provide clear and immediate feedback for all user actions.
- Preference: Ease of use is prioritized over ease of learning.

5.4.4 Performance (Efficiency)

- System response time shall not exceed:
    - **2 seconds** for internal portal operations
    - **3 seconds** for public pages
- Performance degradation shall not exceed **20%** under doubled user load.
- Priority: High

5.4.5 Maintainability

- The system shall follow a modular architecture to allow independent updates.
- Bug fixes or feature updates shall require changes to **no more than 20% of the affected module**.
- Code shall be documented and follow consistent standards.
- Priority: Medium

5.4.6 Security

- The system shall follow secure coding practices and protect against common vulnerabilities (SQL injection, XSS, CSRF).
- All sensitive operations shall require authentication and authorization.
- Priority: Critical

5.5 Business Rules

The following rules define operational constraints and role-based behavior within the system:

- **Role-Based Operations**
  - Supervisors may only mark attendance for workers within their assigned department.
- **Authentication Enforcement**
  - All internal portal pages require authentication.
  - Unauthenticated users shall not access any management functionality.
- **Attendance Workflow**
  - Attendance entries submitted by Supervisors require Admin approval before inclusion in official reports.
  - Approved attendance entries may only be edited by the Admin.
  - All edits shall be recorded in an **audit trail**.
- **Order Processing Workflow**
  - Orders must follow a strict stage sequence:
    - New → In Production → Quality Check → Packed → Dispatched → Delivered
  - Skipping or reversing stages is not permitted.
- **Inquiry Management**
  - Customer inquiries must be acknowledged by the Admin before they can be:
    - closed, or
    - converted into orders
- **Data Ownership**
  - All website assets (source code, database, media, documents) remain the sole property of **Mittal Spinners**.
- **Confidentiality**
  - The development team shall maintain strict confidentiality of all business and operational data.

6. Other Requirements

6.1 Error Handling Requirements

- REQ-ERR-1: System shall log every application error with: error code, module, timestamp, user ID (if authenticated), and a plain-text error message. Logs shall be written to a rotating daily log file with 30-day retention.
- REQ-ERR-2: System shall display user-friendly error messages for all recoverable errors. Messages shall: (a) state what went wrong in plain language; (b) include the error code for support reference; (c) suggest a corrective action. Stack traces shall not be displayed to end users.
- REQ-ERR-3: System shall handle the following edge cases gracefully without crashing:
  - Inquiry form submission with missing mandatory fields: display inline validation errors.
  - Duplicate attendance entry attempt: display ERR-ATT-001 with the existing record date.
  - Invalid order stage transition: display ERR-ORD-002 with allowed transitions.

  - SMTP failure during email notification: log the failure; continue processing the primary action (attendance save / order update) without blocking it.
  - Database connection failure: display ERR-DB-001 and a maintenance message on the appropriate page.
- REQ-ERR-4: System shall provide an Admin-accessible error log viewer showing the last 200 application errors, filterable by date range and module.

Error Code	Module	Description
ERR-WEB-001	Public Website	Invalid or incomplete inquiry form submission
ERR-AUTH-001	Authentication	Invalid credentials supplied
ERR-AUTH-002	Authentication	Session expired – re-login required
ERR-ATT-001	Attendance	Attendance already marked for this worker on this date
ERR-ATT-002	Attendance	Invalid attendance status value
ERR-ORD-001	Order Management	Duplicate order reference detected
ERR-ORD-002	Order Management	Invalid order stage transition
ERR-NOT-001	Notifications	Email delivery failure – SMTP error
ERR-DB-001	Database	Database connection failure
WARN-ATT-001	Attendance	Bulk import contains rows with missing worker IDs
WARN-INQ-001	Inquiry	Inquiry already converted to an order

6.2 Data Integrity Requirements

- REQ-INT-1: Through database-level foreign key constraints for table relationships, the system will guarantee referential integrity. It is best to avoid deleting parent records that have references to child records (no cascading delete). Employees who have a history of attendance should be soft removed (designated as inactive).
- REQ-INT-2: System shall enforce the following database-level CHECK constraints:
  - ATTENDANCE.status IN ('Present', 'Absent', 'Half-day', 'Leave')
  - ATTENDANCE.approval_status IN ('Pending', 'Approved', 'Rejected')
  - ORDER.stage IN ('New', 'In Production', 'Quality Check', 'Packed', 'Dispatched', 'Delivered')
  - INQUIRY.status IN ('New', 'In Progress', 'Resolved', 'Converted')
- REQ-INT-3: Every timestamp field (created_at, updated_at, approved_at, etc.) must be saved in UTC and shown in the user interface in IST (UTC+5:30).
- REQ-INT-4: A supervisor cannot change approved attendance entries. The application (service) layer, not only the user interface, will refuse any attempt at modification made by a non-Admin user.
- REQ-INT-5: All employee records and customer inquiry data will never be permanently removed from the database. The only method of removal that is allowed is soft deletion (status flag).

Appendix A: Glossary

Term / Abbreviation	Definition
Admin	The primary system user with full access to all modules including product management, order tracking, attendance approval, and content updates.
Attendance Module	The internal web module used by Supervisors to mark daily worker attendance and by Admins to approve and generate reports.
CMS (Content Management)	A set of Admin-only pages allowing updates to homepage banners, company info, and product details without developer assistance.
Corporate Website	The public-facing website representing Mittal Spinners, displaying product catalogue, manufacturing process, and contact details.
Department	An operational unit within the Mittal Spinners factory (e.g., Carding, Spinning, Packing) to which workers are assigned.
Inquiry	A customer request submitted via the website contact form, stored in the database and visible to the Admin for follow-up.
Order	A confirmed customer purchase request managed through defined stages from creation to delivery.
Order Stage	One of the defined progress states of an order: New, In Production, Quality Check, Packed, Dispatched, Delivered.
RBAC	Role-Based Access Control – a security model assigning system permissions to roles (Admin, Supervisor) rather than individual users.
Session Timeout	Automatic logout after a period of inactivity to protect authenticated sessions.
Shift	A defined work period (e.g., Morning, Evening, Night) to which workers are assigned for attendance tracking.
SRS	Software Requirements Specification – this document, defining all functional and non-functional requirements for the Mittal Spinners web system.
Supervisor	A mid-level system user responsible for marking daily attendance for workers in their assigned department.
Worker	A factory employee whose attendance is tracked in the system. Workers do not have system login access.
Yarn Catalogue	The product listing section of the public website, displaying cotton yarn types with technical specifications and downloadable catalogues.

Appendix C: To Be Determined List

The following items are pending clarification and will be finalized in future revisions of the SRS:

1. Final deployment environment details (cloud provider and hosting configuration).
2. Exact email service provider (e.g., SMTP service) for notification system.
3. Maximum expected number of concurrent users for scalability planning.
4. Backup and disaster recovery strategy (frequency and storage location).
5. Detailed report formats and customization requirements for PDF and Excel exports.

6. Final UI/UX design specifications and branding elements for the public website and internal portal.
7. Third-party integrations, if any, to be included in future system versions.